NASA's Chandra Releases New 3D Models of Cosmic Objects - NASA

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New three-dimensional (3D) models of objects in space have been released by NASA's Chandra X-ray Observatory. These 3D models allow people to explore — and print — examples of stars in the early and end stages of their lives. They also provide scientists with new avenues to investigate scientific questions and find insights about the objects they represent.

These 3D models are based on state-of-the-art theoretical models, computational algorithms, and observations from space-based telescopes like Chandra that give us accurate pictures of these cosmic objects and how they evolve over time.

However, looking at images and animations is not the only way to experience this data. The four new 3D printable models of Cassiopeia A (Cas A), G292.0+1.8 (G292), Cygnus Loop supernova remnants, and the star known as BP Tau let us experience the celestial objects in the form of physical structures that will allow anyone to hold replicas of these stars and their surroundings and examine them from all angles.

Using NASA's James Webb Space Telescope, astronomers uncovered a mysterious feature within the remnant, nicknamed the "Green Monster," alongside a puzzling network of ejecta filaments forming a web of oxygen-rich material. When combined with X-rays from Chandra, the data helped astronomers shed light on the origin of the Green Monster and revealed new insights into the explosion that created Cas A about 340 years ago, from Earth's perspective.

This 3D model shows a star less than 10 million years old that is surrounded by a disk of material. This class of objects is known as T Tauri stars, named after a young star in the Taurus star-forming region. The model describes the effects of multiple flares, or outbursts that are detected in X-rays by Chandra from one T Tauri star known as BP Tau. These flares interact with the disk of material and lead to the formation of an extended outer atmosphere composed by hot loops, connecting the disk to the developing star.

The Cygnus Loop (also known as the Veil Nebula) is a supernova remnant, the remains of the explosive death of a massive star. This 3D model is the result of a simulation describing the interaction of a blast wave from the explosion with an isolated cloud of the interstellar medium (that is, dust and gas in between the stars). Chandra sees the blast wave and other material that has been heated to millions of degrees. The Cygnus Loop is a highly extended, but faint, structure on the sky: At three degrees across, it has the diameter of six full moons.

This is a rare type of supernova remnant observed to contain large amounts of oxygen. The X-ray image of G292.0+1.8 from Chandra shows a rapidly expanding, intricately structured field left behind by the shattered star. By creating a 3D model of the system, astronomers have been able to examine the asymmetrical shape of the remnant that can be explained by a "reverse" shock wave moving back toward the original explosion.

The 3D models here are the subject of several scholarly papers by Salvatore Orlando of INAF in Palermo, Italy, and colleagues published in The Astrophysical Journal, Astronomy & Astrophysics, and Monthly Notices of the Royal Astronomical Society. Much of this work is also publicly available work on SketchFab.

NASA's Marshall Space Flight Center manages the Chandra program. The Smithsonian Astrophysical Observatory's Chandra X-ray Center controls science operations from Cambridge,

Massachusetts, and flight operations from Burlington, Massachusetts.

Read more from NASA's Chandra X-ray Observatory.

Learn more about the Chandra X-ray Observatory and its mission here:

https://www.nasa.gov/chandra

https://chandra.si.edu

This release features visualizations of three supernova remnants and one star. Each is rendered as a composite image, and as a digital 3-dimensional model, presented in separate short video clips. The composite images are two dimensional and static, but the digital models rotate, showcasing their three-dimensionality.

The first featured supernova is Cassiopeia A. In the X-ray, optical, and infrared composite image, the debris from an exploded star resembles a round purple gas cloud, marbled with streaks of golden light. In the rotating, 3D model, the purple gas cloud is depicted as a flat disk, like a record or CD. Bursting out the front and back of the disk is an orange and white shape similar to a ball of coral, or a head of cauliflower lined with stubby tendrils. Most of the ball, and the majority of the tendrils, appear on one side of the disk. On the opposite side, the shape resembles dollops of thick whipped cream.

Next in the release is a star known as BP Tau. BP Tau is a developing star, less than 10 million years old, and prone to outbursts or flares. These flares interact with a disk of material that surrounds the young star, forming hot loops of extended atmosphere. In the composite image, BP Tau resembles a distant, glowing white dot surrounded by a band of pink light. The rotating, 3D model is far more dynamic and intriguing! Here, the disk of material resembles a large blue puck with round, ringed, concave surfaces. At the heart of the puck is a small, glowing red orb: the developing star. Shooting out of the orb are long, thin, green strands: the flares. Also emerging from the orb are orange and pink petal-shaped blobs: the loops of extended atmosphere. Together, the orb, strands, and petals resemble an exotic flowering orchid.

The third celestial object in this release is the supernova remnant called Cygnus Loop. In the composite image, the remnant resembles a wispy cloud in oranges, blues, purples, and whites, shaped like a backwards letter C. The 3D model examines this cloud of interstellar material interacting with the superheated, supernova blast wave. In the 3D model, the Cygnus Loop resembles a bowl with a thick base, and a wedge cut from the side like a slice of pie. The sides of the bowl are rendered in swirled blues and greens. However, inside the thick base, revealed by the wedge-shaped cut, are streaks of red and orange. Surrounding the shape are roughly parallel thin red strands, which extend beyond the top and bottom of the digital model.

The final supernova featured in this release is G292.0+1.8. The composite image depicts the remnant as a bright and intricate ball of red, blue, and white X-ray gas and debris set against a backdrop of gleaming stars. In the 3D model, the remnant is rendered in translucent icy blue and shades of orange. Here, the rotating shape is revealed to be somewhat like a bulbous arrowhead, or perhaps an iceberg on its side.

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